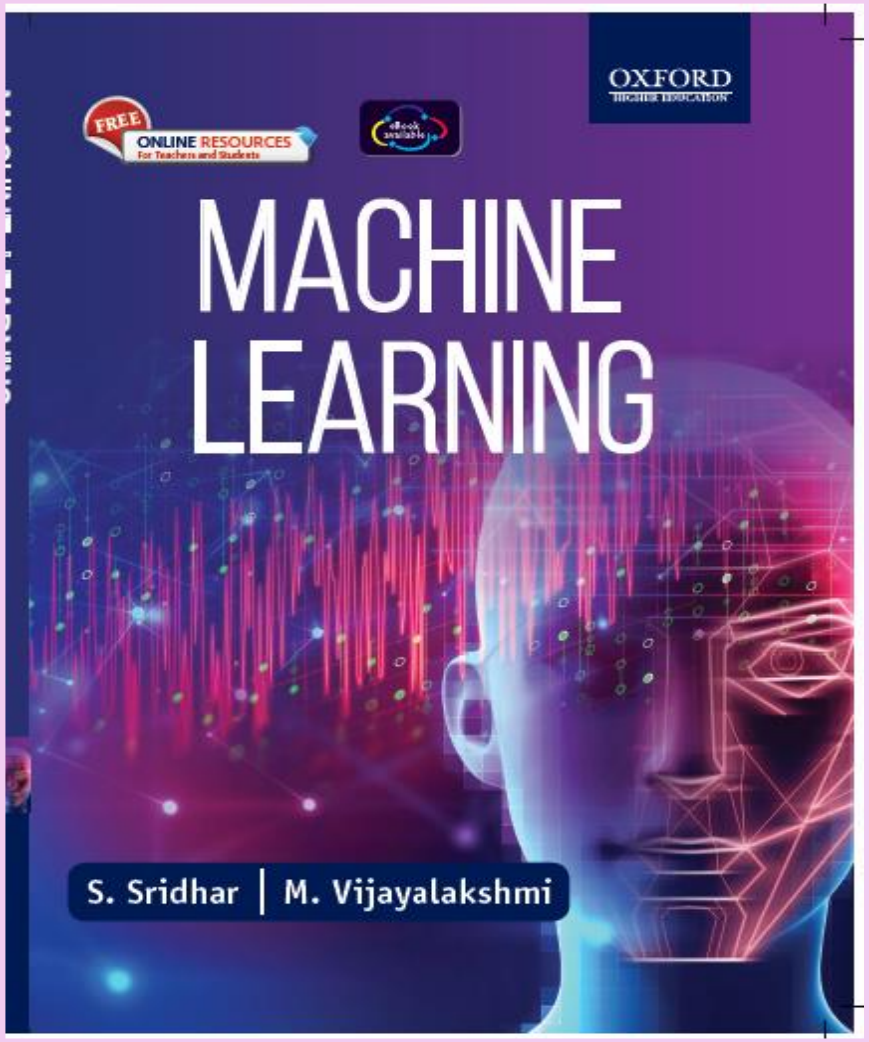


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Machine Learning

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Chapter 6

Decision Tree Learning

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Decision Tree Learning Model

Decision tree learning model, one of the most popular supervised predictive learning models, classifies data instances with high accuracy and consistency. The model performs an *inductive inference* that reaches a general conclusion from observed examples. This model is variably used for solving complex classification applications.

Decision tree is a concept tree which summarizes the information contained in the training dataset in the form of a tree structure. Once the concept model is built, test data can be easily classified.

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Structure of a Decision Tree

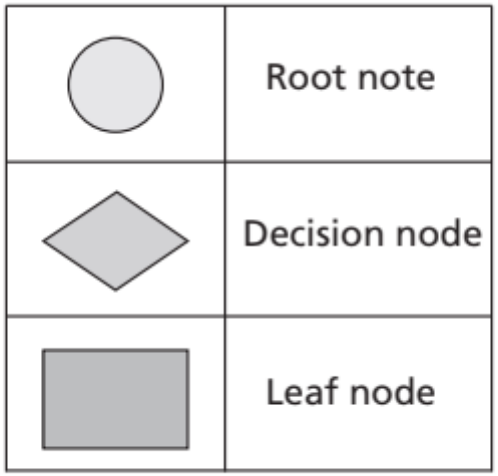


Figure 6.1: Nodes in a Decision Tree

Note: *A decision tree is not always a binary tree. It is a tree which can have more than two branches.*

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Structure of a Decision Tree

BUILDING THE TREE

Goal Construct a decision tree with the given training dataset. The tree is constructed in a top-down fashion. It starts from the root node. At every level of tree construction, we need to find the best split attribute or best decision node among all attributes. This process is recursive and continued until we end up in the last level of the tree or finding a leaf node which cannot be split further. The tree construction is complete when all the test conditions lead to a leaf node. The leaf node contains the target class or output of classification.

Output Decision tree representing the complete hypothesis space.

KNOWLEDGE INFERENCE OR CLASSIFICATION

Goal Given a test instance, infer to the target class it belongs to.

Classification Inferring the target class for the test instance or object is based on inductive inference on the constructed decision tree. In order to classify an object, we need to start traversing the tree from the root. We traverse as we evaluate the test condition on every decision node with the test object attribute value and walk to the branch corresponding to the test's outcome. This process is repeated until we end up in a leaf node which contains the target class of the test object.

Output Target label of the test instance.

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Advantages of Decision Trees

1. Easy to model and interpret
2. Simple to understand
3. The input and output attributes can be discrete or continuous predictor variables.
4. Can model a high degree of nonlinearity in the relationship between the target variables and the predictor variables
5. Quick to train

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Disadvantages of Decision Trees

Some of the issues that generally arise with a decision tree learning are that:

1. It is difficult to determine how deeply a decision tree can be grown or when to stop growing it.
2. If training data has errors or missing attribute values, then the decision tree constructed may become unstable or biased.
3. If the training data has continuous valued attributes, handling it is computationally complex and has to be discretized.
4. A complex decision tree may also be over-fitting with the training data.
5. Decision tree learning is not well suited for classifying multiple output classes.
6. Learning an optimal decision tree is also known to be NP-complete.

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Fundamentals of Entropy

Entropy is the amount of uncertainty or randomness in the outcome of a random variable or an event. Moreover, entropy describes about the homogeneity of the data instances. The best feature is selected based on the entropy value. For example, when a coin is flipped, head or tail are the two outcomes, hence its entropy is lower when compared to rolling a dice which has got six outcomes.

Higher the entropy → Higher the uncertainty
Lower the entropy → Lower the uncertainty

Let P be the probability distribution of data instances from 1 to n as shown in Eq. (6.2).

So, $P = P_1 \ldots\ldots P_n$ (6.2)

Entropy of P is the information measure of this probability distribution given in Eq. (6.3),

$$\text{Entropy_Info}(P) = \text{Entropy_Info}(P_1 \ldots P_n)$$
$$= -(P_1 \log_2(P_1) + P_2 \log_2(P_2) + \ldots\ldots + P_n \log_2(P_n)) \quad (6.3)$$

where, P_1 is the probability of data instances classified as class 1 and P_2 is the probability of data instances classified as class 2 and so on.

P_1 = |No of data instances belonging to class 1| / |Total no of data instances in the training dataset|

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Algorithm 6.1: General Algorithm for Decision Trees

Algorithm 6.1: General Algorithm for Decision Trees

1. Find the best attribute from the training dataset using an attribute selection measure and place it at the root of the tree.
2. Split the training dataset into subsets based on the outcomes of the test attribute and each subset in a branch contains the data instances or tuples with the same value for the selected test attribute.
3. Repeat step 1 and step 2 on each subset until we end up in leaf nodes in all the branches of the tree.
4. This splitting process is recursive until the stopping criterion is reached.

Stopping Criteria

The following are some of the common stopping conditions:

1. The data instances are homogenous which means all belong to the same class C_i and hence its entropy is 0.
2. A node with some defined minimum number of data instances becomes a leaf (Number of data instances in a node is between 0.25 and 1.00% of the full training dataset).
3. The maximum tree depth is reached, so further splitting is not done and the node becomes a leaf node.

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Decision Tree Induction Algorithms

There are many decision tree algorithms, such as ID3, C4.5, CART, CHAID, QUEST, GUIDE, CRUISE, and CTREE, that are used for classification in real-time environment. The most commonly used decision tree algorithms are ID3 (Iterative Dichotomizer 3), developed by J.R Quinlan in 1986, and C4.5 is an advancement of ID3 presented by the same author in 1993. CART, that stands for Classification and Regression Trees, is another algorithm which was developed by Breiman et al. in 1984.

The accuracy of the tree constructed depends upon the selection of the best split attribute. Different algorithms are used for building decision trees which use different measures to decide on the splitting criterion. Algorithms such as ID3, C4.5 and CART are popular algorithms used in the construction of decision trees. The algorithm ID3 uses 'Information Gain' as the splitting criterion whereas the algorithm C4.5 uses 'Gain Ratio' as the splitting criterion. The CART algorithm is popularly used for classifying both categorical and continuous-valued target variables. CART uses GINI Index to construct a decision tree.

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ID3 Tree Construction

Algorithm 6.2: Procedure to Construct a Decision Tree using ID3

1. Compute Entropy_Info Eq. (6.8) for the whole training dataset based on the target attribute.
2. Compute Entropy_Info Eq. (6.9) and Information_Gain Eq. (6.10) for each of the attribute in the training dataset.
3. Choose the attribute for which entropy is minimum and therefore the gain is maximum as the best split attribute.
4. The best split attribute is placed as the root node.
5. The root node is branched into subtrees with each subtree as an outcome of the test condition of the root node attribute. Accordingly, the training dataset is also split into subsets.
6. Recursively apply the same operation for the subset of the training set with the remaining attributes until a leaf node is derived or no more training instances are available in the subset.

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C4.5 Construction

Algorithm 6.3: Procedure to Construct a Decision Tree using C4.5

1. Compute Entropy_Info Eq. (6.8) for the whole training dataset based on the target attribute.
2. Compute Entropy_Info Eq. (6.9), Info_Gain Eq. (6.10), Split_Info Eq. (6.11) and Gain_Ratio Eq. (6.12) for each of the attribute in the training dataset.
3. Choose the attribute for which Gain_Ratio is maximum as the best split attribute.
4. The best split attribute is placed as the root node.
5. The root node is branched into subtrees with each subtree as an outcome of the test condition of the root node attribute. Accordingly, the training dataset is also split into subsets.
6. Recursively apply the same operation for the subset of the training set with the remaining attributes until a leaf node is derived or no more training instances are available in the subset.

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Dealing with Continuous Attributes in C4.5

The C4.5 algorithm is further improved by considering attributes which are continuous, and a continuous attribute is discretized by finding a split point or threshold. When an attribute '*A*' has numerical values which are continuous, a threshold or best split point '*s*' is found such that the set of values is categorized into two sets such as $A < s$ and $A \geq s$. The best split point is the attribute value which has maximum information gain for that attribute.

Now, let us consider the set of continuous values for the attribute CGPA in the sample dataset as shown in Table 6.12.

Table 6.12: Sample Dataset

S.No.	CGPA	Job Offer
1.	9.5	Yes
2.	8.2	Yes
3.	9.1	No
4.	6.8	No
5.	8.5	Yes
6.	9.5	Yes
7.	7.9	No
8.	9.1	Yes
9.	8.8	Yes
10.	8.8	Yes

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Dealing with Continuous Attributes in C4.5

First, sort the values in an ascending order.

6.8	7.9	8.2	8.5	8.8	8.8	9.1	9.1	9.5	9.5

Remove the duplicates and consider only the unique values of the attribute.

6.8	7.9	8.2	8.5	8.8	8.8	9.1	9.5

Now, compute the Gain for the distinct values of this continuous attribute. Table 6.13 shows the computed values.

Table 6.13: Gain Values for CGPA

	6.8		7.9		8.2		8.5		8.8		9.1		9.5	
Range	≤	>	≤	>	≤	>	≤	>	≤	>	≤	>	≤	>
Yes	0	7	0	7	1	6	2	5	4	3	5	2	7	0
No	1	2	2	1	2	1	2	1	2	1	3	0	3	0
Entropy	0	0.7637	0	0.5433	0.9177	0.5913	1	0.6497	0.9177	0.8108	0.9538	0	0.8808	0
Entropy_Info (*S*, *T*)	0.6873		0.4346		0.6892		0.7898		0.8749		0.7630		0.8808	
Gain	0.1935		**0.4462**		0.1916		0.091		0.0059		0.1178		0	

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Dealing with Continuous Attributes in C4.5

Table 6.14: Discretized Instances

S.No.	CGPA Continuous	CGPA Discretized	Job Offer
1.	9.5	>7.9	Yes
2.	8.2	>7.9	Yes
3.	9.1	>7.9	No
4.	6.8	≤7.9	No
5.	8.5	>7.9	Yes
6.	9.5	>7.9	Yes
7.	7.9	≤7.9	No
8.	9.1	>7.9	Yes
9.	8.8	>7.9	Yes
10.	8.8	>7.9	Yes

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Classification and Regression Trees Construction

Algorithm 6.4: Procedure to Construct a Decision Tree using CART

1. Compute Gini_Index Eq. (6.13) for the whole training dataset based on the target attribute.
2. Compute Gini_Index for each of the attribute Eq. (6.14) and for the subsets of each attribute in the training dataset.
3. Choose the best splitting subset which has minimum Gini_Index for an attribute.
4. Compute ΔGini Eq. (6.15) for the best splitting subset of that attribute.
5. Choose the best splitting attribute that has maximum ΔGini.
6. The best split attribute with the best split subset is placed as the root node.
7. The root node is branched into two subtrees with each subtree an outcome of the test condition of the root node attribute. Accordingly, the training dataset is also split into two subsets.
8. Recursively apply the same operation for the subset of the training set with the remaining attributes until a leaf node is derived or no more training instances are available in the subset.

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Regression Trees

Regression trees are a variant of decision trees where the target feature is a continuous valued variable. These trees can be constructed using an algorithm called reduction in variance which uses standard deviation to choose the best splitting attribute.

Algorithm 6.5: Procedure for Constructing Regression Trees

1. Compute standard deviation for each attribute with respect to target attribute.
2. Compute standard deviation for the number of data instances of each distinct value of an attribute.
3. Compute weighted standard deviation for each attribute.
4. Compute standard deviation reduction by subtracting weighted standard deviation for each attribute from standard deviation of each attribute.
5. Choose the attribute with a higher standard deviation reduction as the best split attribute.
6. The best split attribute is placed as the root node.
7. The root node is branched into subtrees with each subtree as an outcome of the test condition of the root node attribute. Accordingly, the training dataset is also split into different subsets.
8. Recursively apply the same operation for the subset of the training set with the remaining attributes until a leaf node is derived or no more training instances are available in the subset.

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Validating and Pruning of Decision Trees

To avoid overfitting of the tree, we need to prune the trees and construct an optimal decision tree. Trees can be pre-pruned or post-pruned. If tree nodes are pruned during construction or the construction is stopped earlier without exploring the nodes' branches, then it is called as pre-pruning whereas if tree nodes are pruned after the construction is over then it is called as post-pruning. Basically, the dataset is split into three sets called training dataset, validation dataset and test dataset. Generally, 40% of the dataset is used for training the decision tree and the remaining 60% is used for validation and testing. Once the decision tree is constructed, it is validated with the validation dataset and the misclassifications are identified. Using the number of instances correctly classified and number of instances wrongly classified, Average Squared Error (ASE) is computed. The tree nodes are pruned based on these computations and the resulting tree is validated until we get a tree that performs better. Cross validation is another way to construct an optimal decision tree. Here, the dataset is split into k-folds, among which k–1 folds are used for training the decision tree and the kth fold is used for validation and errors are computed. The process is repeated for randomly k–1 folds and the mean of the errors is computed for different trees. The tree with the lowest error is chosen with which the performance of the tree is improved. This tree can now be tested with the test dataset and predictions are made.

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Validating and Pruning of Decision Trees

Some of the tree pruning methods are listed below:

1. Reduced Error Pruning
2. Minimum Error Pruning (MEP)
3. Pessimistic Pruning
4. Error–based Pruning (EBP)
5. Optimal Pruning
6. Minimum Description Length (MDL) Pruning
7. Minimum Message Length Pruning
8. Critical Value Pruning

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Summary

1. The decision tree learning model performs an *Inductive inference* that reaches a general conclusion from observed examples.
2. The decision tree learning model generates a complete hypothesis space in the form of a tree structure.
3. A decision tree has a structure that consists of a root node, internal nodes/decision nodes, branches, and terminal nodes/leaf nodes.
4. Every path from root to a leaf node represents a logical rule that corresponds to a conjunction of test attributes and the whole tree represents a disjunction of these conjunctions.
5. A decision tree consists of two major procedures, namely building the tree and knowledge inference or classification.
6. A decision tree is constructed by finding the attribute or feature that best describes the target class for the given test instances.

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Summary

7. Decision tree algorithms such as ID3, C4.5, CART, CHAID, QUEST, GUIDE, CRUISE, and CTREE, are used for classification.
8. The univariate decision tree algorithm ID3 uses 'Information Gain' as the splitting criterion whereas algorithm C4.5 uses 'Gain Ratio' as the splitting criterion.
9. Multivariate decision tree algorithm called Classification and Regression Trees (CART) algorithm is popularly used for classifying both categorical and continuous-valued target variables.
10. CART uses GINI Index to construct a decision tree.
11. ID3 works well if the attributes or features are considered as discrete/categorical values.
12. C4.5 and CART can handle both categorical attributes and continuous attributes.
13. Both C4.5 and CART can also handle missing values. C4.5 is prone to outliers but CART can handle outliers too.
14. The C4.5 algorithm is further improved by considering attributes which are continuous, and a continuous attribute is discretized by finding a split point or threshold.
15. The algorithm C4.5 is based on Occam's Razor which says that given two correct solutions, the simpler solution must be chosen.
16. Regression trees are a variant of decision trees where the target feature is a continuous-valued variable.

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